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Customer Lifetime Value (CLV) Prediction

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Course Project- PMDS5040L C1+TC1

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Dataset Selection and Problem Statement

The dataset used in this study is an e-commerce transactional dataset containing approximately 500,000 transaction-level records. Each record includes CustomerID, InvoiceDate, Quantity, UnitPrice, and product details. The dataset was preprocessed by removing null CustomerIDs, filtering out negative quantities and unit prices, and converting invoice dates to datetime format. A revenue column was derived as the product of Quantity and UnitPrice.

For regression analysis, the dataset was aggregated at the customer level, yielding approximately 3,000–5,000 observations. A time-based train/test split was applied at the 75th percentile of InvoiceDate to preserve temporal ordering. The dependent variable (CLV) was defined as the total revenue a customer generates in the 90 days following the training period's end date.

Independent Variables (Feature Set)

- Recency — Days since the customer's last purchase
- Frequency — Number of unique invoices
- Monetary Total — Total revenue generated
- Average Order Value (AOV) — Mean revenue per transaction
- Product Diversity — Number of unique products purchased
- Tenure — Days between first and last purchase

- Purchase Velocity — Frequency divided by (Tenure + 1)

The objective was to apply regression techniques to predict CLV and quantify the relationship between customer behavioral features and future revenue.

Model Specification

A multiple linear regression model was specified as the primary analytical tool:

$$\text{CLV} = \beta_0 + \beta_1(\text{Recency}) + \beta_2(\text{Frequency}) + \beta_3(\text{Monetary}) + \beta_4(\text{AOV}) + \beta_5(\text{Diversity}) + \beta_6(\text{Tenure}) + \beta_7(\text{Velocity}) + \varepsilon$$

Linear regression was selected because it provides interpretable coefficients for each feature, supports statistical hypothesis testing (t-tests and F-tests), and establishes a transparent baseline against which more complex models can be benchmarked.

Regression Assumptions

- Linearity between each predictor and the CLV response
- Independence of error terms across observations
- Homoscedasticity — constant variance of residuals
- Normality of residuals for valid inference

Parameter Estimation

The coefficients for the initial model were estimated using the Ordinary Least Squares (OLS) method via the statsmodels library. The results are shown below.

Initial Model Summary (OLS on raw CLV)

Variable	Coefficient	Std. Error	t-statistic	P-value
const	140.29	170.99	0.820	0.412
recency	-0.07	0.75	-0.098	0.922
frequency	32.48	14.41	2.253	0.024
monetary_total	0.18	0.01	17.253	0.000
avg_order_value	-0.75	0.47	-1.581	0.114
product_diversity	2.20	1.11	1.983	0.047
tenure	-1.70	0.86	-1.989	0.047
purchase_velocity	-23.84	135.07	-0.177	0.860

Interpretation:

- Intercept (140.29): The expected CLV when all predictor variables are zero. This baseline value is not statistically significant ($p = 0.412$).
- Frequency (32.48): Each additional purchase transaction is associated with an increase of approximately 32 units in CLV. This indicates that higher engagement customers generate significantly more future revenue.

- **Monetary Total (0.1766):** For every one-unit increase in total historical spending, CLV increases by 0.18 units. This is the strongest and most reliable predictor in the model ($p < 0.001$).
- **Product Diversity (2.20):** Customers who purchase across a wider range of product categories tend to have higher CLV — each additional unique product category is associated with a 2.20-unit increase.
- **Tenure (-1.70):** A negative coefficient suggests that longer-tenured customers may have reached spending saturation, or that their spending pattern is already reflected in other variables.
- **Average Order Value (-0.75) and Purchase Velocity (-23.84):** Both are statistically insignificant ($p > 0.1$), suggesting limited incremental predictive value in this model.

4. Model Significance and Goodness of Fit

4.1 Individual Variable Significance (t-tests)

Each coefficient is tested with $H_0: \beta_i = 0$ against $H_a: \beta_i \neq 0$.

Variable	p-Value	Decision at $\alpha = 0.05$
Recency	0.922	Fail to reject H_0
Frequency	0.024	Reject H_0 ✓
Monetary Total	<0.001	Reject H_0 ✓

Avg Order Value	0.114	Fail to reject H_0
Product Diversity	0.047	Reject H_0 ✓
Tenure	0.047	Reject H_0 ✓
Purchase Velocity	0.860	Fail to reject H_0

4.2 Overall Model Significance (F-test)

H_0 : All coefficients are jointly zero

H_a : At least one coefficient is non-zero

Metric	Value
F-Statistic	86.88
p-value	9.52e-116 (≈ 0)
No. of Observations	3,088
Degrees of Freedom (Model)	7
Degrees of Freedom (Residuals)	3,080

Conclusion: The F-test result is highly significant ($p \approx 0$), meaning the regression model as a whole explains a statistically significant portion of variance in CLV.

4.3 Goodness of Fit

Metric	Value	Interpretation
R²	0.165	16.5% variance explained
Adjusted R²	0.163	Adjusted for 7 predictors
AIC	58,700	Lower = better fit
BIC	58,750	Penalizes complexity
RMSE (Validation)	1,943.14	Average prediction error

While the model is statistically significant, the R^2 of 0.165 indicates weak explanatory power. This is common in real-world CLV datasets due to high variance in individual customer behavior and missing demographic or psychographic features.

5. Residual Analysis and Diagnostic Tests

5.1 Residual Plot

The scatter plot of residuals versus fitted values shows a non-random fan-shaped pattern, with residuals increasing in spread as predicted values grow.

This is a clear visual indicator of heteroscedasticity — a violation of the constant variance assumption.

5.2 Normality of Residuals

The Q-Q (Quantile-Quantile) probability plot reveals severe departures from the theoretical normal line, especially in the tails. Heavy skewness is apparent (Skewness = 46.07, Kurtosis = 2368.09), driven by extreme outlier customers with very high CLV values.

Jarque-Bera Test: $p\text{-value} = 0.00 \rightarrow$ Strongly reject normality.

5.3 Homoscedasticity Test (Breusch-Pagan)

The Breusch-Pagan test formally evaluates H_0 : Constant error variance vs. H_a : Heteroscedasticity present.

Test	Statistic / p-value	Result
Breusch-Pagan	$p = 2.08 \times 10^{-156}$	Strong heteroscedasticity ✗
Jarque-Bera	$p = 0.000$	Non-normal residuals ✗
Durbin-Watson	≈ 2.001	No autocorrelation ✓

5.4 Multicollinearity

The condition number is 2.51×10^4 , which is well above the threshold of 1,000.

This suggests the presence of moderate-to-strong multicollinearity among the predictors, which can inflate standard errors and reduce the reliability of individual coefficient estimates.

5.5 Summary of Assumption Checks

Assumption	Status	Evidence
Linearity	VIOLATED	Non-random residual pattern
Normality of Residuals	VIOLATED	JB test $p = 0$, heavy skewness
Homoscedasticity	VIOLATED	BP $p = 2.08 \times 10^{-156}$
Independence (No Autocorrelation)	SATISFIED	Durbin-Watson ≈ 2.001

No Multicollinearity	MILD CONCERN	Condition No. = 25,100
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6. Model Improvement

6.1 Motivation for Transformation

Given the severe violations of normality and homoscedasticity in the original model — largely driven by the right-skewed distribution of CLV — a logarithmic transformation of the target variable was applied. Log transformations are a standard technique for compressing the range of highly skewed positive outcomes.

6.2 Log-Transformed Model

The transformation applied:

$$\text{CLV_log} = \log(1 + \text{CLV})$$

This form ($\log(1+x)$) handles cases where $\text{CLV} = 0$ without producing undefined values.

6.3 Log Model Regression Results

Variable	Coefficient	Std. Error	t-Stat	p-Value
Intercept	1.9123	0.147	13.043	<0.001
Recency	-0.0029	0.001	-4.581	<0.001
Frequency	0.0511	0.012	4.134	<0.001
Monetary Total	2.41e-05	8.78e-06	2.743	0.006
Avg Order Value	-0.0004	0.000	-0.996	0.319
Product Diversity	0.0052	0.001	5.503	<0.001
Tenure	0.0057	0.001	7.734	<0.001
Purchase Velocity	-0.1747	0.116	-1.509	0.132

Key change: Recency now becomes a significant predictor ($p < 0.001$, negative sign), indicating that more recently active customers are associated with higher future CLV. Additionally, Tenure's sign reverses to positive in the log model, suggesting a compounding loyalty effect when outliers are controlled.

6.4 Model Comparison

Metric	Original Model	Log- Transformed Model	Improvement
R²	0.165	0.215	+30.3% relative gain
Adjusted R²	0.163	0.213	Better fit
F-Statistic	86.88	120.5	More significant
RMSE (Validation)	1,943.14	1,399.12	-28% error reduction
Skewness	46.07	0.29	Near normal
Kurtosis	2,368.09	1.97	Normal range
Residual Normality	Severely violated	Substantially improved	Yes

6.5 Alternative Model: Random Forest

A Random Forest Regressor with 200 estimators was also trained as a benchmark non-linear model. Despite its ability to capture complex interactions and non-linearities, the model underperformed the log-transformed linear regression:

Model	RMSE	Verdict
Log-Transformed OLS	1,399.12	Best performer ✓
Random Forest (200 trees)	1,704.48	Higher error
Original OLS	1,943.14	Worst performer

This result suggests that after proper variable transformation, a well-specified linear model can outperform ensemble methods — particularly when the sample size is moderate and interpretability is valued.

7. Business Insights and Segmentation

7.1 Revenue Concentration (Pareto Analysis)

Top 20% of customers contribute 76.53% of total predicted revenue.

This finding mirrors the classic Pareto Principle and has major implications for resource allocation in customer retention programs. Marketing budgets should be disproportionately allocated to this high-value segment.

7.2 Key Predictors and Business Implications

Feature	Direction	Business Insight
Purchase Frequency	Positive	Reward repeat buyers — loyalty programs and reorder reminders
Monetary Total	Positive	High spenders drive CLV — offer premium tiers or exclusive perks
Product Diversity	Positive	Cross-sell promotions increase CLV — bundle recommendations
Recency (Log Model)	Negative	Recent customers are more valuable — activate lapsing customers quickly
Tenure	Mixed	Long-term loyalty matters — but older customers may be saturating

7.3 Recommended Business Actions

- **High-Frequency Customers:** Target with loyalty programs, personalized offers, and subscription incentives to sustain engagement.
- **Cross-Category Shoppers:** Identify single-category customers and introduce bundle discounts or recommendation algorithms to increase product diversity.

- At-Risk High-Value Customers: Detect early lapse signals through recency monitoring and deploy win-back campaigns before customers churn.
- VIP Segment (Top 20%): Create a dedicated high-value customer program with premium support, early access to products, and tailored offers.

8. Conclusion

8.1 Summary of Findings

This study demonstrates the application of multiple linear regression to predict Customer Lifetime Value from RFM-based behavioral features. Key findings include:

- Monetary Total is the single strongest predictor of CLV ($p < 0.001$), confirming that past spending is the best indicator of future value.
- Purchase Frequency and Product Diversity are significant secondary drivers, aligning with established CLV theory.
- Log transformation of CLV substantially improved model diagnostics: R^2 increased from 0.165 to 0.215, and RMSE dropped by 28%.

- The log-linear model outperformed Random Forest (RMSE: 1,399 vs. 1,704), demonstrating the power of proper preprocessing over model complexity.
- Top 20% of customers account for 76.53% of predicted revenue — a strong case for differential customer treatment strategies.

8.2 Limitations

- Low R^2 (16.5–21.5%) indicates significant unexplained variance, likely due to missing variables such as customer demographics, browsing behavior, and promotional response history.
- Persistent multicollinearity (condition number = 25,100) may inflate standard errors and reduce reliability of individual coefficient estimates.
- The model assumes linearity even after transformation, which may not fully capture complex, non-linear relationships.
- The 90-day prediction window may not capture long-term customer dynamics.

8.3 Future Improvements

- Feature Engineering: Include customer demographics, geographic data, channel preferences, and seasonal patterns.
- Advanced Modeling: Test XGBoost, LightGBM, and deep learning architectures for potentially improved predictive power.

- Multicollinearity Remediation: Apply Variance Inflation Factor (VIF) screening or Principal Component Analysis (PCA) to decorrelate features.
- Probabilistic CLV: Implement BG/NBD + Gamma-Gamma models for a statistically grounded probabilistic CLV framework.
- Cross-Validation: Replace single train/test split with k-fold cross-validation for more robust evaluation.

9. Appendix: Python Code

```
[1]: import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error

from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor

import statsmodels.api as sm
from statsmodels.stats.diagnostic import het_breuschpagan
import scipy.stats as stats

np.random.seed(42)

[2]: df = pd.read_csv(R"C:\Users\Neharika\Downloads\data.csv", encoding='ISO-8859-1')

df = df.dropna(subset=['CustomerID'])
df = df[df['Quantity'] > 0]
df = df[df['UnitPrice'] > 0]

df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])
df['Revenue'] = df['Quantity'] * df['UnitPrice']
df['CustomerID'] = df['CustomerID'].astype(str)

[3]: split_date = df['InvoiceDate'].quantile(0.75)

train_df = df[df['InvoiceDate'] <= split_date]

[4]: prediction_window_days = 90
reference_date = train_df['InvoiceDate'].max()

future_df = df[(df['InvoiceDate'] > reference_date) & (df['InvoiceDate'] <=
reference_date + pd.Timedelta(days=prediction_window_days))]
```

```
future_revenue = future_df.groupby('CustomerID')['Revenue'].sum().reset_index()
future_revenue.columns = ['CustomerID', 'CLV']
```

```
5]: def create_features(data, reference_date):
```

```
    customer_df = data.groupby('CustomerID').agg({'InvoiceDate': ['max',
↳ 'min'], 'InvoiceNo': 'nunique', 'Revenue': ['sum', 'mean'], 'StockCode':
↳ 'nunique'}).reset_index()

    customer_df.columns = ['CustomerID', 'last_purchase', 'first_purchase',
↳ 'frequency', 'monetary_total', 'avg_order_value', 'product_diversity']

    customer_df['recency'] = (reference_date - customer_df['last_purchase']).dt.
↳ days
    customer_df['tenure'] = (customer_df['last_purchase'] -
↳ customer_df['first_purchase']).dt.days
    customer_df['purchase_velocity'] = customer_df['frequency'] /
↳ (customer_df['tenure'] + 1)

    return customer_df
```

```
6]: reference_date_train = train_df['InvoiceDate'].max()
train_features = create_features(train_df, reference_date_train)

model_df = train_features.merge(future_revenue, on='CustomerID', how='left')
model_df['CLV'] = model_df['CLV'].fillna(0)
```

```
2]:
```

```
7]: features =
↳ ['recency', 'frequency', 'monetary_total', 'avg_order_value', 'product_diversity', 'te

X = model_df[features]
y = model_df['CLV']

X_train, X_val, y_train, y_val = train_test_split(X, y, test_size=0.2,
↳ random_state=42)

X_train_sm = sm.add_constant(X_train)
X_val_sm = sm.add_constant(X_val)

8]: model = sm.OLS(y_train, X_train_sm).fit()
print(model.summary())
```

OLS Regression Results

OLS Regression Results

Dep. Variable:	CLV	R-squared:	0.165
Model:	OLS	Adj. R-squared:	0.163
Method:	Least Squares	F-statistic:	86.88
Date:	Tue, 31 Mar 2026	Prob (F-statistic):	9.52e-116
Time:	21:29:28	Log-Likelihood:	-29343.
No. Observations:	3088	AIC:	5.870e+04
Df Residuals:	3080	BIC:	5.875e+04
Df Model:	7		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	140.2923	170.994	0.820	0.412	-194.982	475.567
recency	-0.0735	0.748	-0.098	0.922	-1.541	1.393
frequency	32.4776	14.413	2.253	0.024	4.218	60.737
monetary_total	0.1766	0.010	17.253	0.000	0.157	0.197
avg_order_value	-0.7487	0.473	-1.581	0.114	-1.677	0.180
product_diversity	2.1988	1.109	1.983	0.047	0.025	4.372
tenure	-1.7025	0.856	-1.989	0.047	-3.381	-0.024
purchase_velocity	-23.8406	135.067	-0.177	0.860	-288.670	240.989

Omnibus:	9522.015	Durbin-Watson:	2.001
Prob(Omnibus):	0.000	Jarque-Bera (JB):	720808648.575
Skew:	46.066	Prob(JB):	0.00
Kurtosis:	2368.090	Cond. No.	2.51e+04

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 2.51e+04. This might indicate that there are strong multicollinearity or other numerical problems.

```

: y_pred = model.predict(X_val_sm)
rmse = np.sqrt(mean_squared_error(y_val, y_pred))

print("Original RMSE:", rmse)

```

Original RMSE: 1943.140516876309

```

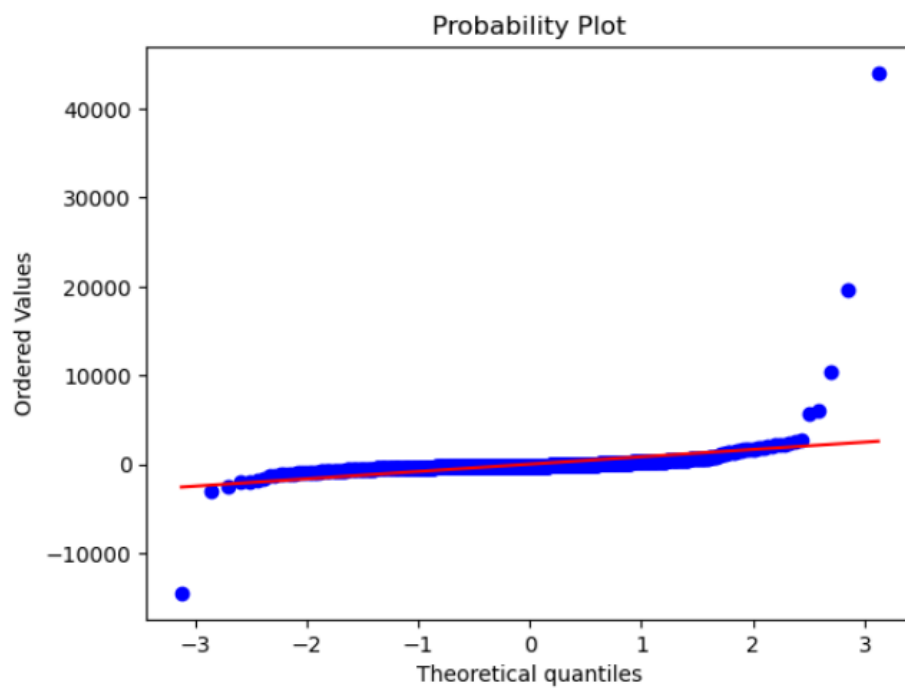
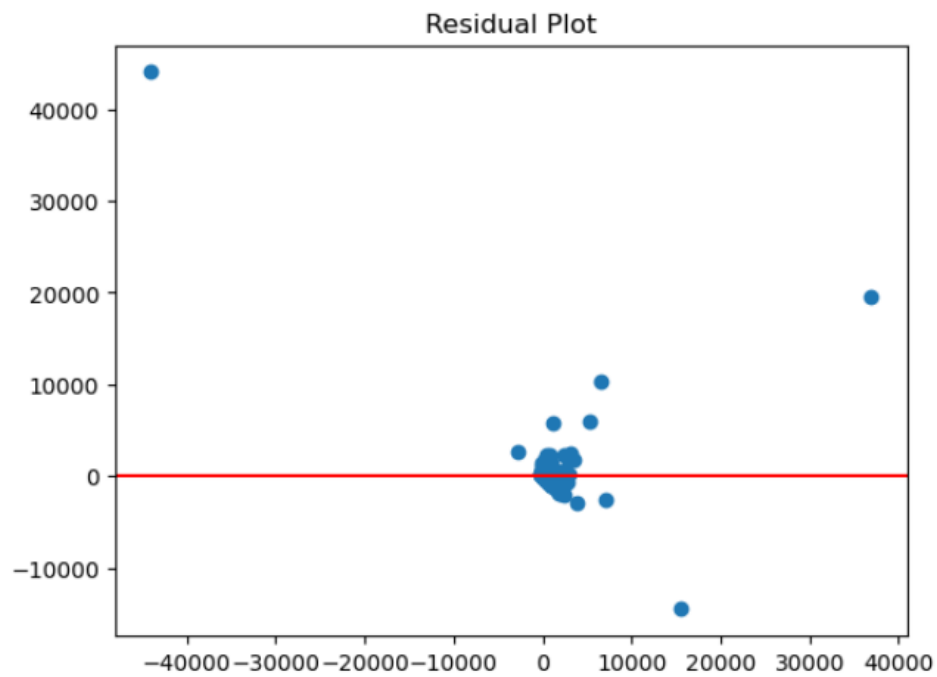
]: residuals = y_val - y_pred

# Residual Plot
plt.scatter(y_pred, residuals)
plt.axhline(0, color='red')
plt.title("Residual Plot")
plt.show()

# Q-Q Plot
stats.probplot(residuals, dist="norm", plot=plt)
plt.show()

# Breusch-Pagan Test
bp_test = het_breuschpagan(residuals, X_val_sm)
print("BP p-value:", bp_test[1])

```



BP p-value: 2.0809375575160206e-156

```
[11]: # Log transform
model_df['CLV_log'] = np.log1p(model_df['CLV'])

y_log = model_df['CLV_log']

X_train, X_val, y_train, y_val = train_test_split(X, y_log, test_size=0.2,
→random_state=42)

X_train_sm = sm.add_constant(X_train)
X_val_sm = sm.add_constant(X_val)

[12]: model_log = sm.OLS(y_train, X_train_sm).fit()
print(model_log.summary())
```

```

-----
                        OLS Regression Results
=====
Dep. Variable:          CLV_log      R-squared:                0.215
Model:                  OLS          Adj. R-squared:           0.213
Method:                 Least Squares   F-statistic:             120.5
Date:                  Tue, 31 Mar 2026   Prob (F-statistic):       7.23e-157
Time:                  21:31:02         Log-Likelihood:          -7536.6
No. Observations:      3088            AIC:                    1.509e+04
Df Residuals:          3080            BIC:                    1.514e+04
Df Model:               7
Covariance Type:       nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
const	1.9123	0.147	13.043	0.000	1.625	2.200
recency	-0.0029	0.001	-4.581	0.000	-0.004	-0.002
frequency	0.0511	0.012	4.134	0.000	0.027	0.075
monetary_total	2.407e-05	8.78e-06	2.743	0.006	6.86e-06	4.13e-05
avg_order_value	-0.0004	0.000	-0.996	0.319	-0.001	0.000
product_diversity	0.0052	0.001	5.503	0.000	0.003	0.007
tenure	0.0057	0.001	7.734	0.000	0.004	0.007
purchase_velocity	-0.1747	0.116	-1.509	0.132	-0.402	0.052

```
=====
Omnibus:                 861.664    Durbin-Watson:           2.048
Prob(Omnibus):            0.000    Jarque-Bera (JB):        180.521
Skew:                     0.292    Prob(JB):                6.32e-40
Kurtosis:                 1.969    Cond. No.:               2.51e+04
=====
```

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 2.51e+04. This might indicate that there are strong multicollinearity or other numerical problems.

```
y_pred_log = model_log.predict(X_val_sm)

# Convert back
y_pred = np.expm1(y_pred_log)
y_true = np.expm1(y_val)

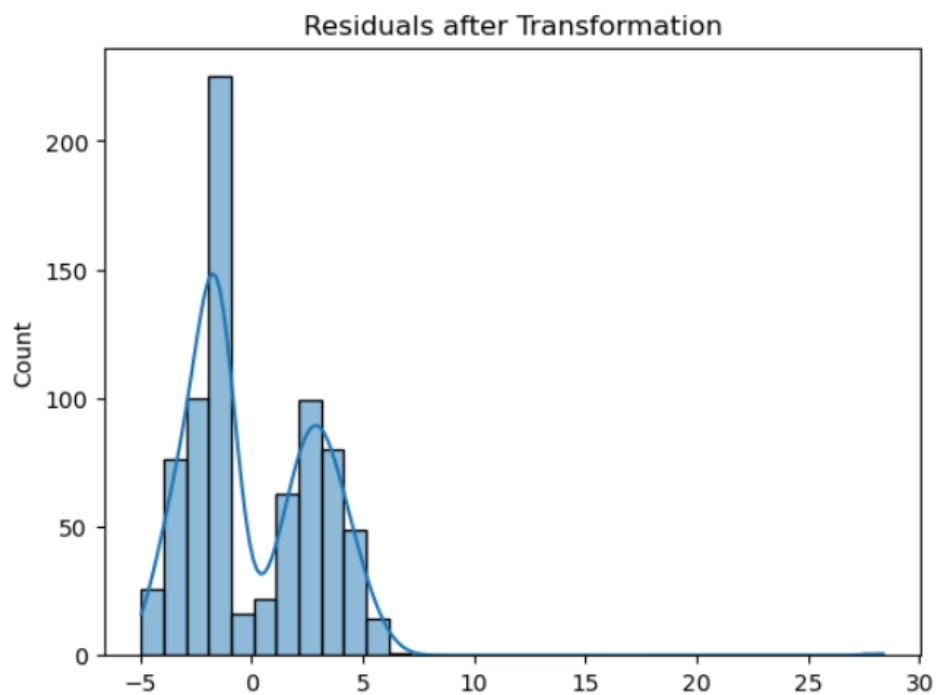
rmse_log = np.sqrt(mean_squared_error(y_true, y_pred))

print("RMSE after log transform:", rmse_log)

RMSE after log transform: 1399.1226401063714
```

```
residuals_log = y_val - y_pred_log

sns.histplot(residuals_log, kde=True)
plt.title("Residuals after Transformation")
plt.show()
```



```
: rf = RandomForestRegressor(n_estimators=200, random_state=42, n_jobs=-1)
```

```
rf.fit(X_train, np.expml(y_train))
```

```
rf_pred = rf.predict(X_val)
```

```
rf_rmse = np.sqrt(mean_squared_error(y_true, rf_pred))
```

```
print("Random Forest RMSE:", rf_rmse)
```

Random Forest RMSE: 1704.4781835636359

```
: top_20 = model_df.nlargest(int(0.2 * len(model_df)), 'predicted_clv')
```

```
revenue_share = top_20['predicted_clv'].sum() / model_df['predicted_clv'].sum()
```

```
print(f"Top 20% customers contribute {revenue_share:.2%} of predicted revenue")
```

Top 20% customers contribute 76.53% of predicted revenue